

Predictive Structure Framework for Dynamical Systems: An Information-Based Approach to Predictive Representation Selection

Predictive Structure Project

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Abstract

A dynamical system does not necessarily possess a single fixed predictive representation. Instead, predictive structure depends on the prediction horizon, available information, representation space, and observer constraints. This paper introduces the Predictive Structure Framework (PSF), which formally defines an optimal predictive representation as the one that maximizes accessible predictive information under a temporal constraint: $\Phi^*(h) = \arg \max_{\Phi} I(\Phi, h)$. By treating machine learning models as measurement devices with inherent inductive biases, the framework separates physical predictive accessibility from observer-induced representation preference, offering an operational criterion for tracking how predictive structure appears, changes, and becomes inaccessible over time.

1 Introduction

Complex dynamical systems can often be represented in multiple ways, and different representations may become useful at different prediction horizons. A critical question arises: How can we determine which representation carries the true predictive structure of the system?

While machine learning models can discover useful representations, model-selected features often reflect the architecture, optimization method, and inductive bias of the observer rather than physical reality. Therefore, an observer-independent criterion is required. We propose the Predictive Structure Framework (PSF) to bridge this gap.

2 Theoretical Framework

2.1 Predictive Representation

Let Φ be a candidate representation of a dynamical system. The predictive accessibility $I(\Phi, h)$ measures the amount of information available for future prediction at horizon h .

2.2 Optimal Representation

The optimal predictive representation is defined as:

$$\Phi^*(h) = \arg \max_{\Phi} I(\Phi, h) \tag{1}$$

2.3 Representation Regimes

The framework identifies distinct regimes of predictive information dynamics:

- **Regime I (Representation Migration):** The dominant predictive carrier changes domains (e.g., $\Phi_A \rightarrow \Phi_B$).

- **Regime II (Representation Refinement):** The domain remains the same, but the mathematical description improves.
- **Regime III (Information Horizon Collapse):** Predictive information becomes inaccessible.

2.4 Observer Independence Principle

A model may select a representation because it is convenient for its architecture, not because it is physically predictive. To solve this, we introduce the Information-Conditioned Carrier Score (ICCS):

$$ICCS(\Phi, h) = PIS(\Phi, h) \times AO(\Phi, h) \quad (2)$$

which combines the Predictive Information Share (PIS) and Algorithmic Observability (AO) to separate predictive accessibility from observer preference.

3 Experiments

The framework was validated across a suite of dynamical systems.

3.1 Physical Representation Migration

In the Lorenz and Rössler systems, we observed Representation Migration. As the prediction horizon increases, local coordinate predictability collapses, forcing the optimal representation to migrate to cross-variable or phase-based representations.

3.2 Observer-Induced Migration

In the Van der Pol oscillator, we tested observer-induced bias. Algorithms with distinct inductive biases (e.g., Random Forests) hallucinated structural transitions that were not physically present. Applying the ICCS metric filtered out these algorithmic hallucinations, demonstrating the robustness of the Observer Independence Principle.

4 Limitations

The framework explicitly bounds its claims:

- Predictive information accessibility is not strictly synonymous with physical causality.
- An optimal predictive representation does not guarantee the discovery of a fundamental physical mechanism.
- The framework is bounded by the completeness of the candidate representation space.

Future work, organized as Boundary Stress Tests, will explore hidden variables, causal/predictive divergence, and adversarial representation spaces.

5 Conclusion

The Predictive Structure Framework provides an operational method for analyzing how predictive structure behaves in dynamical systems. By acknowledging that predictive representation is dynamic and observer-dependent, PSF offers a rigorous mathematical approach for identifying true structural transitions versus algorithmic artifacts.

References